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Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 25-03-2026

SUBSCRIPTION-BASED MEMBERSHIP ENGINE WITH TIERED LOGIC

SDG: SDG No – SDG 9 – Industry, Innovation and Infrastructure

Presented By:

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Courses Handling Faculty:

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INTRODUCTION



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- A Subscription-Based Membership Engine with Tiered Logic is a system that manages users, recurring payments, and controlled access to services based on subscription plans.
- It allows businesses to offer multiple membership tiers such as Basic, Standard, and Premium.
- Each tier provides different features, permissions, and benefits.
- The system automatically assigns access rights after successful payment.
- It supports recurring billing, upgrades, downgrades, and cancellations.
- Tiered logic ensures secure and structured feature distribution.
- This model helps generate predictable recurring revenue.
- It is widely used in SaaS platforms, e-learning sites, and streaming services.

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PROBLEM STATEMENT



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➤ Many digital platforms struggle to manage recurring subscriptions and user access efficiently.

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➤ Manual tracking of payments and memberships often leads to errors and revenue loss.

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➤ There is difficulty in controlling feature access based on different subscription levels.

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➤ Upgrades, downgrades, and renewals are not handled smoothly in traditional systems.

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➤ Lack of automated billing causes missed payments and service interruptions.

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➤ Security risks arise when access control is not properly enforced.

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➤ Businesses also face challenges in scaling membership plans.

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➤ Therefore, a structured Subscription-Based Membership Engine with Tiered Logic is needed to automate billing, manage tiers, and control access securely.

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OBJECTIVE(S)



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- To design and develop a scalable Subscription-Based Membership Engine with Tiered Logic that automates recurring billing and securely controls user access based on membership tiers.
- To implement an automated tier-based subscription system that manages recurring payments and dynamically controls user access according to selected membership plans.
- To improve subscription management by enhancing automation, strengthening access control, and increasing scalability and revenue efficiency through structured tiered logic.
- To evaluate performance of of the Subscription-Based Membership Engine by measuring billing accuracy, access control reliability, system scalability, and user satisfaction.

PROPOSED SYSTEM



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- The proposed system is a Subscription-Based Membership Engine with structured tiered access control.

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- It automates user registration, subscription selection, and recurring billing.

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- Multiple membership tiers such as Basic, Standard, and Premium are defined with specific features.

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- The system dynamically grants or restricts access based on the user's active plan.

- It supports secure payment gateway integration for seamless transactions.

- Users can upgrade, downgrade, or cancel subscriptions easily.

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- An admin dashboard allows monitoring of users, plans, and revenue analytics.

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- The system is designed to be scalable, secure, and reliable for growing digital platforms.

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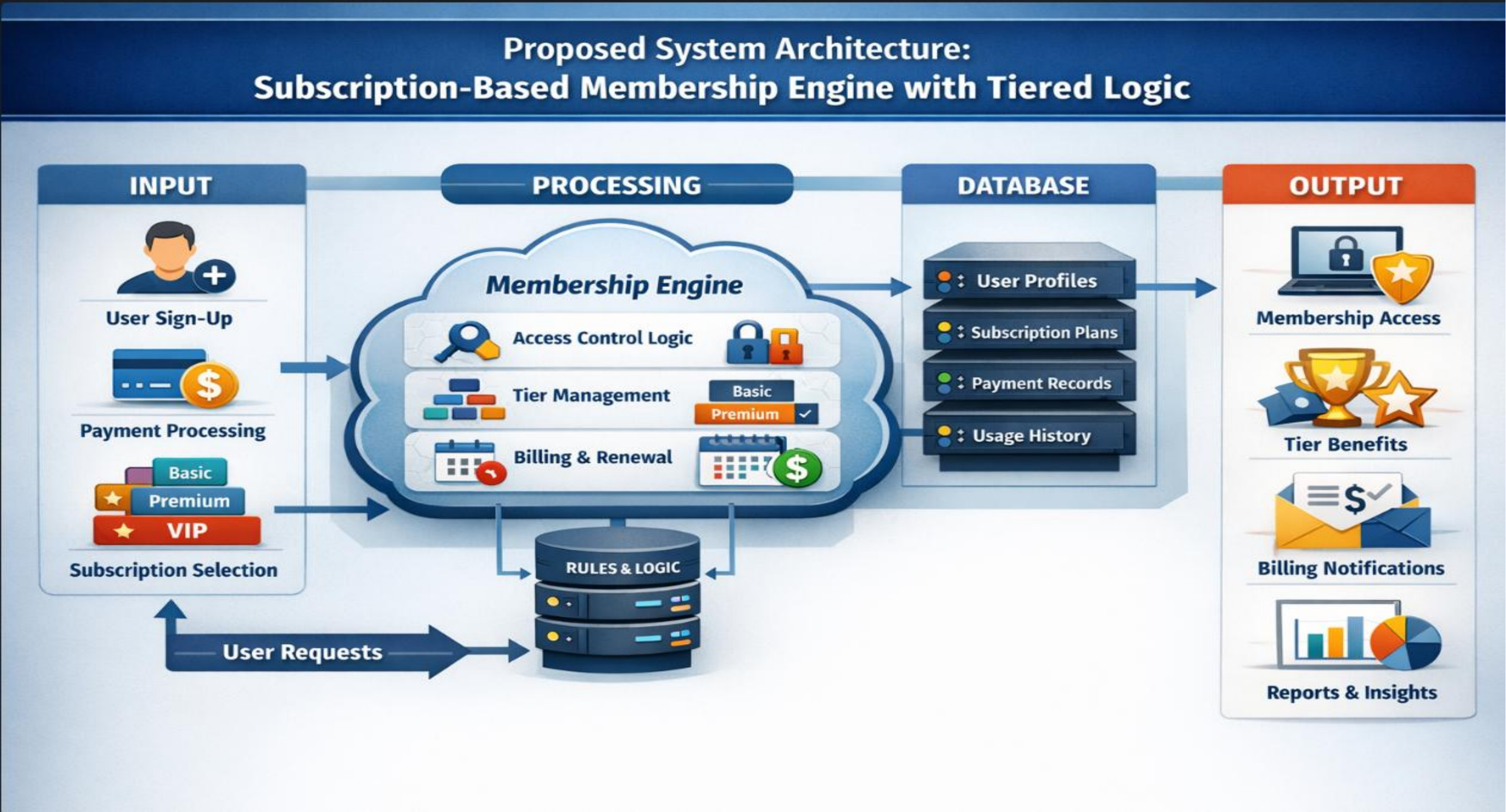
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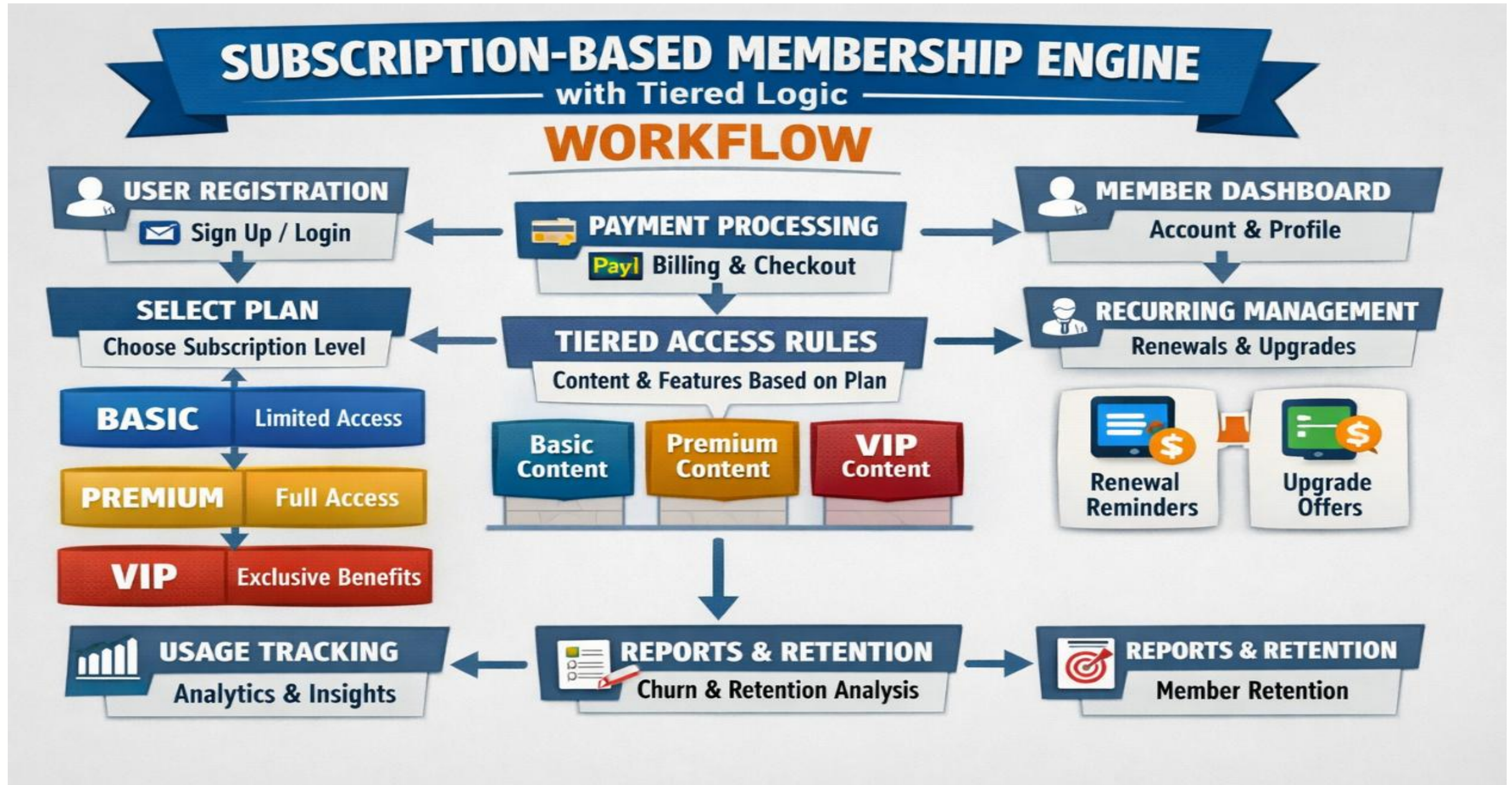
PROPOSED SYSTEM ARCHITECTURE



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WORKFLOW



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TECHNOLOGIES USED



➤ Frontend:

1 HTML5 – Used to design the structure of web pages.

2 CSS3 – Used for styling and responsive design.

➤ Backend:

3 Node.js, Express.js

➤ Database:

4 MySQL

➤ Tools:

5 Visual Studio Code, Git, GitHub

➤ Hardware Requirements:

7 Minimum 4GB RAM, Inter i3 Processor or higher
500GB Hard Disk, Internet Connection

➤ Software Requirements:

8 Windows/Linux/macOS

9 Node.js installed

MangoDB/MySQL installed

Web Browser

Code Editor

MODULE DESCRIPTION



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Module 1: User Management

This module handles all user-related operations such as registration, login, logout, and profile management. It ensures secure authentication using encrypted passwords and token-based authorization.

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Module 2: Core Processing

This module implements the main business logic of the membership engine. It manages subscription plan selection, tier upgrades and downgrades, renewal processing, payment validation, and expiry date calculation.

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Module 3: Database Management

This module manages data storage and retrieval operations. It securely stores user profiles, membership plans, payment records, subscription history, and access permissions.

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Module 4: Reporting & Analytics

This module generates reports and visual insights for administrators. It provides data on active users, revenue by subscription tier, renewal rates, and user growth trends. Analytical dashboards help in tracking system performance and making business decisions. Reports can be exported for further analysis and record-keeping.

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IMPLEMENTATION SCREENSHOTS



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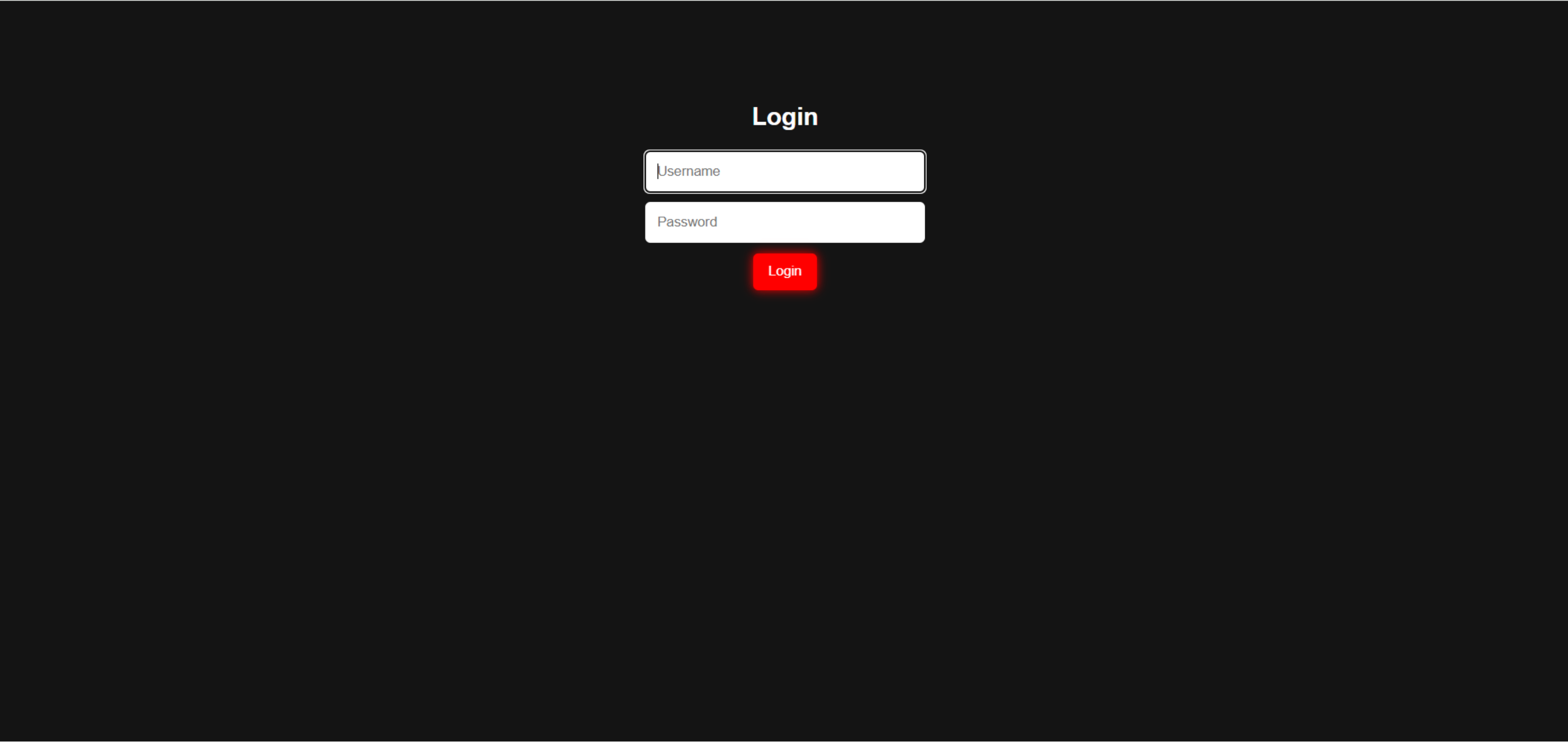
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IMPLEMENTATION SCREENSHOTS



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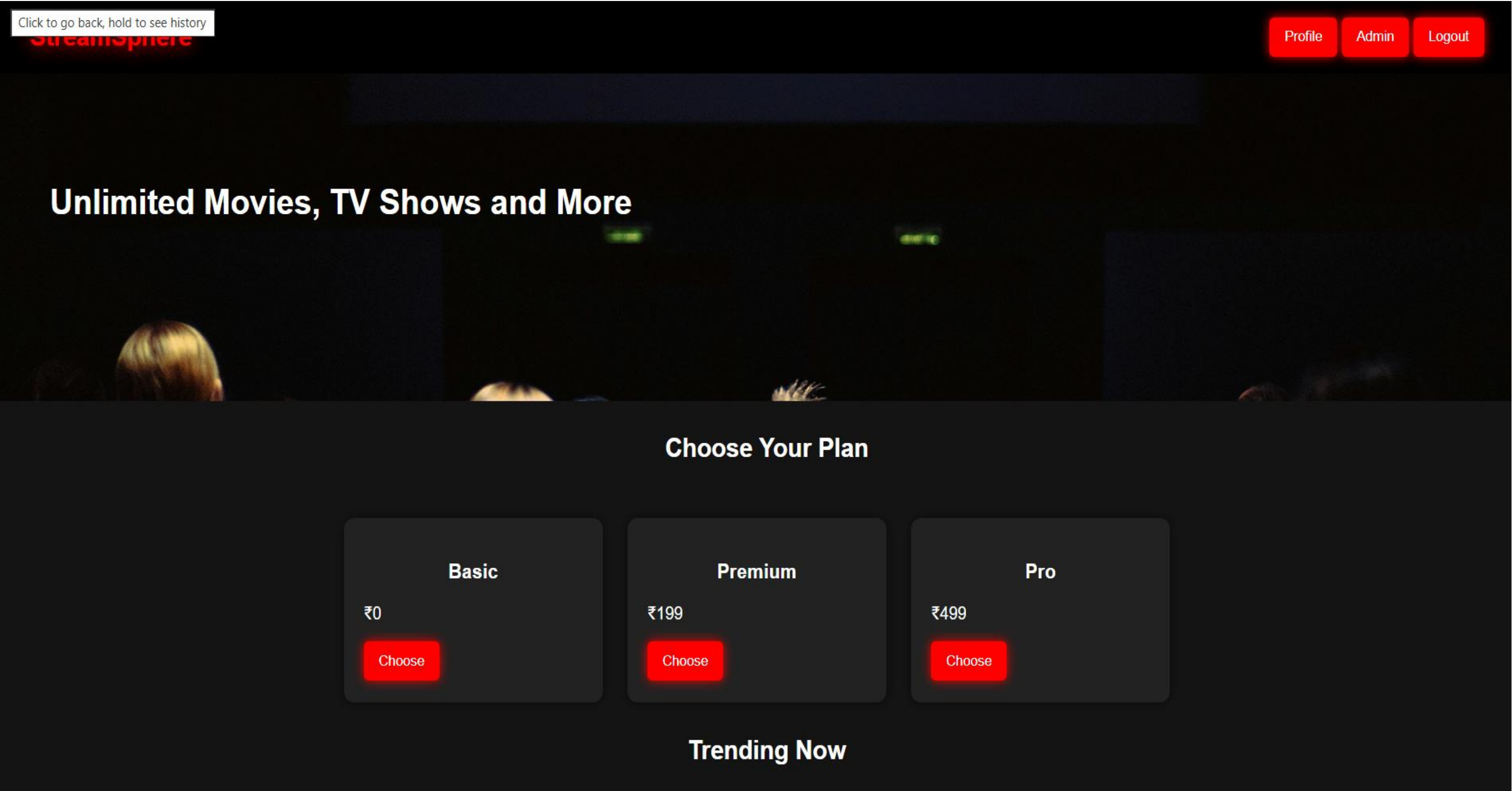
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IMPLEMENTATION SCREENSHOTS



Main Module

Core functionality for managing user subscriptions and membership tiers.

1. User Registration

User Registration

Email

Password

Login

Forgot Password?

3. Select Subscription Plan

Basic	Premium	Enterprise
\$10/month	\$25/month	\$50/month
✓ Limited Access	✓ Full Access	✓ All Features
✓ Email Support	✓ Priority Support	✓ 24/7 Support
Select Plan	Select Plan	Select Plan

4. Payment Processing

Welcome, [User Name]

Current Plan Premium

Expiry Dates 25-06-2024

Status Active

Upgrade Plan Payment History Notifications

4. Payment Processing

Payment

Select Plan Premium - \$25/month

Card Number

Expiry Date CVV

Pay Now

5. Payment Processing

Premium - \$25/month

Card Number

Expiry Date CVV

Pay Now

VISA Mastercard UPI

5. Dashboard & Management

Welcome, [User Name]

Current Plan Premium

Expiry Dates 25-06-2024

Status Active

Upgrade Plan Payment History Notifications

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IMPLEMENTATION SCREENSHOTS



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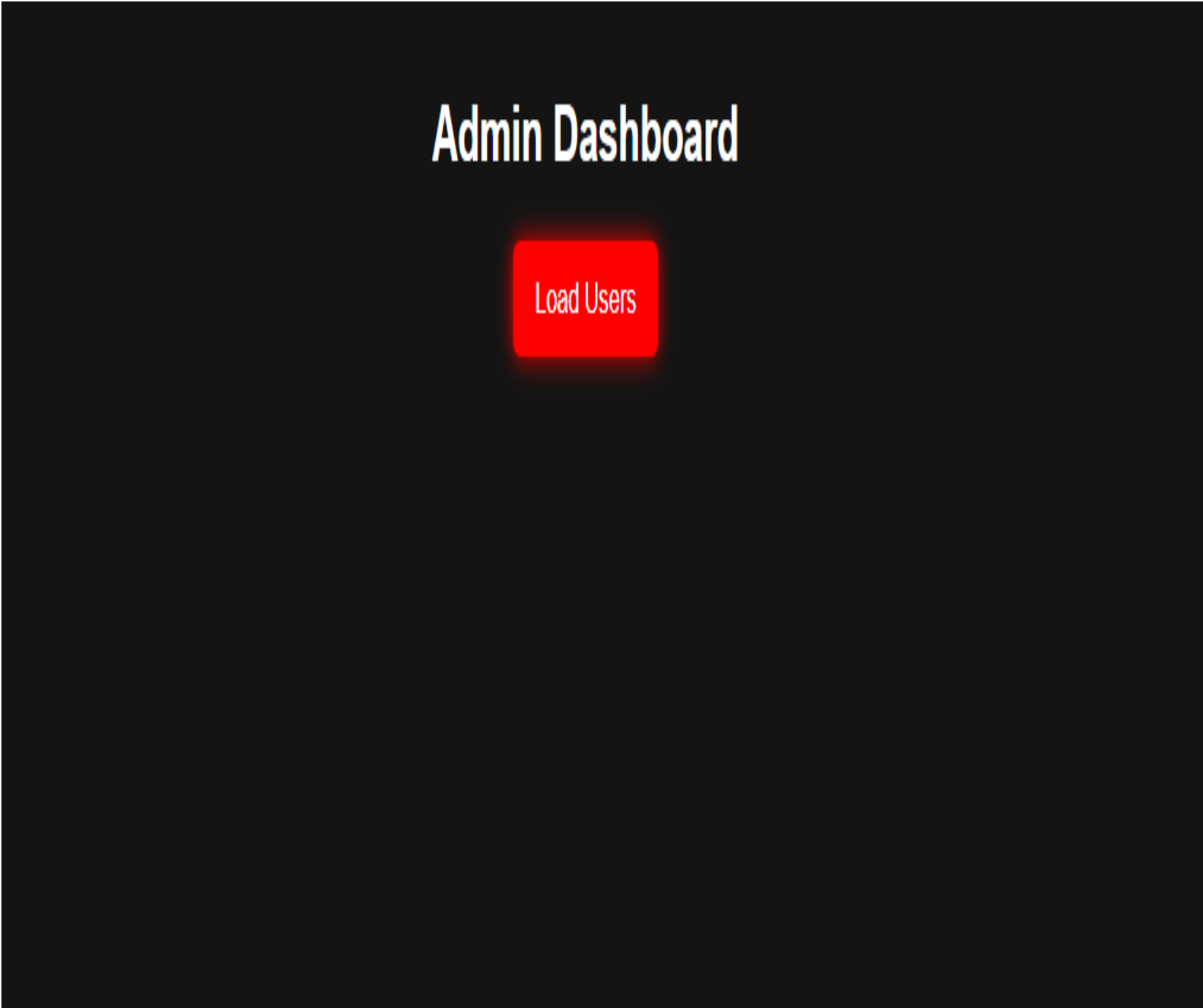
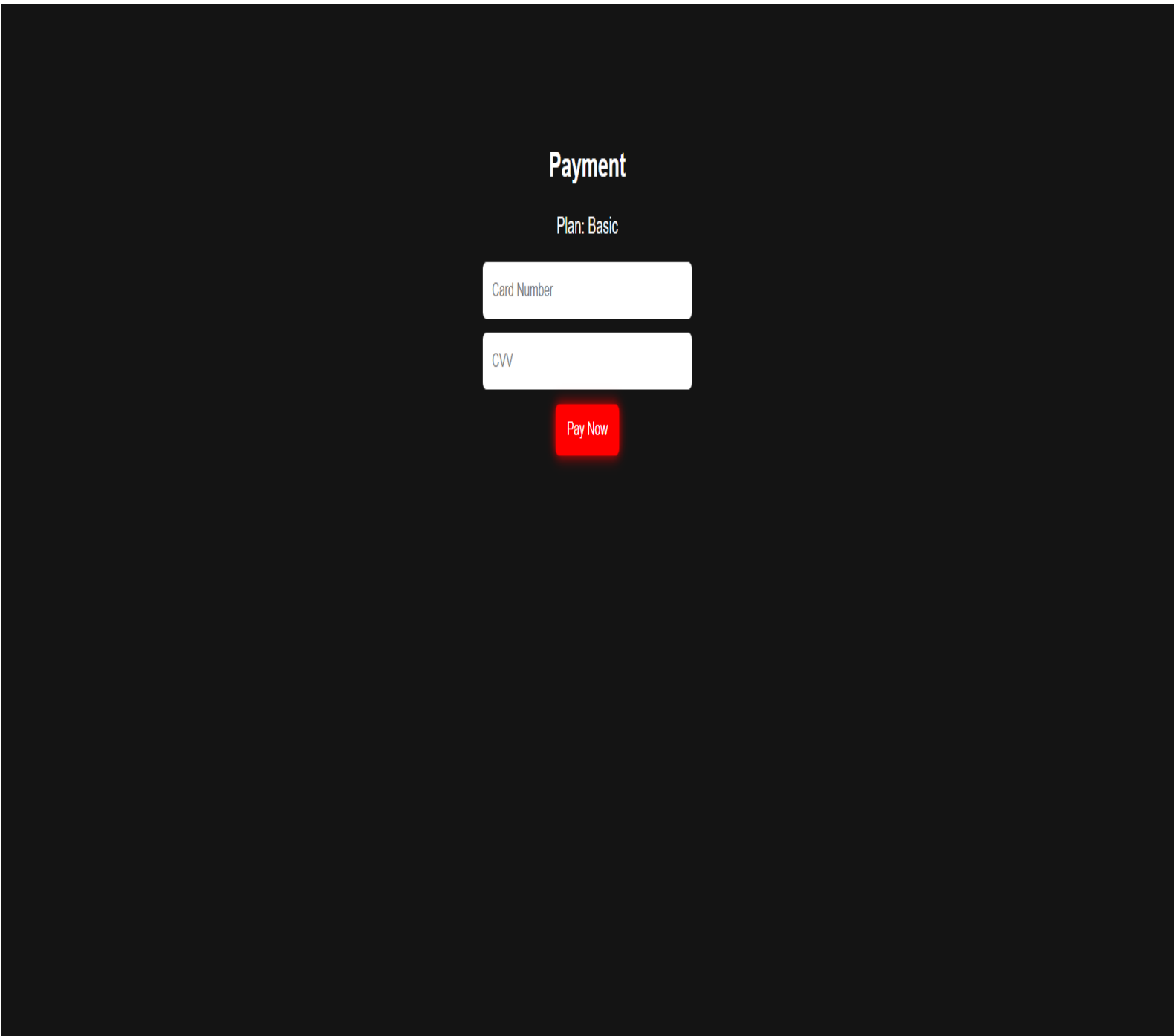
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ADVANTAGES & LIMITATIONS



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Advantages:

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1. Tier-Based Access Control

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2. Scalable Architecture

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3. Automated Subscription Management

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4. Improved Revenue Management

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5. Secure User Authentication

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6. Real-Time Reporting & Analytics

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7. Flexible Plan Customization

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8. Reduced Manual Work

Limitations:

1. Initial Development Cost

2. Dependency on Internet Connectivity

3. Payment Gateway Dependency

4. Data Security Risks

5. Complex Tier Logic Management

6. Maintenance Requirement



Thank you